

16) In 8086, 32-bit addition can be performed by adding low 16-bits of the operand high 16-bits of the operands using _____ instruction.

- A) ADC - SBB
- B) ADD - ADC
- C) ADC - ADC
- D) ADD - ADD
- E) ADC - ADD

17) What is the IEEE 754 floating point equivalent of $-\infty$?

- A) 10000000111111111111111111111111b
- B) 10000000000000000000000000000000b
- C) 00000000000000000000000000000000b
- D) 11111111100000000000000000000000b
- E) 11111111111111111111111111111111b

18) It is given that the following code computes factorial of var_N. Then, what is the value of var_N?
code?

```
MOV AX, 1  
MOV BX, var_N  
MOV CX, var_N
```

label:

```
MUL BX
```

missing instruction?

```
LOOP label
```

```
RET
```

```
var_N DW 6
```

- A) INC AX
- B) DEC CX
- C) INC BX

19) Which field of an instruction specifies the addressing mode to be used?

- A) REG
- B) R/M
- C) Direction

20) Assume that CS=1000H, DS=2000H, SS=3000H, and IP=2000H. The next instruction executed by the microprocessor?

- A) 12000H
- B) 22000H
- C) 1000H

8) Which of the following memory types are volatile?

- I. SRAM
- II. DRAM
- III. Flash
- IV. EPROM

A) I, II, IV

B) III and IV

☒ C) I and II

D) Only IV

☒ E) I, II, III, IV

9) A short jump allows a program to branch forward _____ bytes or backwards _____ bytes from the next instruction's address in the program.

☒ A) 127, 128

B) 128, 127

☒ C) 256, 255

D) 255, 256

~~E) 32K, 32K~~

10) 12-bit Hamming coded data including 8-bit actual data and 4-bit parity is received as given below.

Bit	1	2	3	4	5	6	7	8	9	10	11	12
Data	0	0	1	1	1	0	0	1	0	0	0	0

Using the error checking method provided below, detect position of the error (if any) within the code.

C1 = XOR of bits (1, 3, 5, 7, 9, 11)

C2 = XOR of bits (2, 3, 6, 7, 10, 11)

C4 = XOR of bits (4, 5, 6, 7, 12)

C8 = XOR of bits (8, 9, 10, 11, 12)

A) Bit-2

B) Bit-6

C) No error

☒ D) Bit-10

E) Bit-4

1) Which type of jump instruction is used to jump to any...

- I. Short
- II. Near
- III. Far
- IV. Relative

A) I and II

B) II and III

☒ C) II and IV

☒ D) Only III

E) I and IV

2) How many (128Mx16) memory device is needed to store 1 G bytes of data?

E) 1

☒ C) 4

☒ D) 16

3) INC instruction is a special type of addition. However, _____ flag is not affected by INC instruction unlike ADD.

A) 2

B) 8

E) A

D) P

☒ C) C

4) Regarding SRAM and DRAM, which of the following statements are true?

A) S

B) Z

5) Regarding SRAM and DRAM, which of the following statements are true?

I. DRAM is faster than SRAM.

II. SRAM is more expensive than DRAM.

III. DRAM requires periodical refresh.

IV. SRAM consumes more power than DRAM.

☒ C) I, II, IV

D) I and IV

☒ E) II and III

B) III and IV

E) P

☒ A) Only I

☒ D) O

☒ C) D

6) Which of the following flag bits is not tested by the conditional jump instructions?

B) C

D) NOT

A) S

C) XOR

☒ E) AND

7) Which of the following flags indicates whether the result of an arithmetic operation is positive or negative?

B) NOR

C) O

E) D

A) OR

D) Z

8) Which of the following flags indicates whether the result of an arithmetic operation is positive or negative?

B) C

☒ A) S

Using the error checking methods provided below, detect position of the error

- C1 = XOR of bits (1, 3, 5, 7, 9, 11)
- C2 = XOR of bits (2, 3, 6, 7, 10, 11)
- C4 = XOR of bits (4, 5, 6, 7, 12)
- C8 = XOR of bits (8, 9, 10, 11, 12)

- A) Bit-2
- B) Bit-4
- C) Bit-6
- D)

11) What is the corresponding value of -76.5 in nearest even rounding mode?

- A) -77
- B) -75
- C) -78
- D)

12) The code below computes the sum of numbers from 1 to N using the for instruction should be inserted into the missing line within the code?

$$\sum_{x=1}^N x = N.(N+1)/2$$

```
ORG 100h  
CALL sum  
RET
```

```
sum PROC  
MOV AL, N  
MOV BL, N  
INC BL  
MUL BL  
missing instruction?  
RET  
sum ENDP
```

N DB 8

- A) DEC AL
- B) SHR AL, 1
- C) SUB AL, 2

13) What is the decimal equivalent of IEEE 754 single precision floating point number 0xBF800000?

- A) -20.25
- B) -40.125
- C) -10.50

14) What is the size of Personal Computer I/O map?

4) Which of the following communication architectures are serial?

- I. USB
- II. PCI
- III. PCI Express
- IV. ISA

A) I, II, IV B) I, II, III C) I and III D) Only I

5) Which flag is tested by a conditional interrupt instruction?

A) Interrupt B) Carry C) Parity D) Overflow

6) Regarding SRAM and DRAM, which of the following statements are true?

- I. DRAM is faster than SRAM.
- II. SRAM is more expensive than DRAM.
- III. DRAM requires periodical refresh.
- IV. SRAM consumes more power than DRAM.

A) II and III B) I, II, IV C) Only I D) I and IV

7) What is the IEEE 754 single precision floating point representation of decimal +10.0?

A) 41200000h B) C1A00000h C) 41A00000h D) C1200000h

8) Which of the following statements is not true when an interrupt occurs?

- I. Flags register contents are pushed on the stack.
- II. Interrupt flag is set.
- III. Contents of CS and IP are pushed onto the stack.
- IV. Related ISR is executed.

A) Only II B) I and III C) III and IV D) II and IV

9) Using little endian format, 4321h is to be stored in an 8-bit memory starting at 200 memory address 201h after storing?

A) 34h B) 43h C) 12h D) Empty

1) Instruction Queue is placed in _____ of Intel 8086.

- A) Control and Timing Unit
- B) Execution Unit
- C) Register File
- D) Bus Interface Unit
- E) Arithmetic Logic Unit

2) _____ is the act of synchronizing two systems that op

- A) Serial communication
- B) Handshaking
- C) Transmission
- D) Parallel communication
- E) Peripheral interface

3) Which of the followings are interrupt pins found on Intel micro

- I. INTA
- II. INTC
- III. INTQ
- IV. INTR

- A) I and IV
- B) I, III, IV
- C) Only III

21) Which of the following instructions converts any BCD value available in AX register into ASCII? Hint: ASCII equivalent of a BCD code can be obtained by appending $(011)_2$ to left of the BCD code.

- A) AND AX, 07h
- B) OR AX, 0Fh
- C) AND AX, 0Fh
- D) XOR AX, 0Fh
- E) OR AX, 30h

22) Assume that SP=0100H. What is the value of SP after execution of POP AX instruction?

- A) 00FFH
- B) 0101H
- C) 0102H
- D) 00FEH

23) Which of the following communication architectures are parallel?

- I. USB
- II. PCI
- III. PCI Express
- IV. ISA

- A) I and II
- B) I, II, III
- C) I and III
- D) II and IV

24) How many (128Mx16) memory device is needed to store 1 G bytes of data?

- A) 1
- B) 16
- C) 2
- D) 8

25) Which of the following connections are common to all memory devices?

- I. Address
- II. Read Control
- III. Write Control
- IV. Data

- A) I, II, IV
- B) I, II, III, IV
- C) Only IV
- D) I, II